

Ivan Pavlov

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EDUCATION

EPFL, Lausanne, Switzerland

Sep. 2024 – Present

Master in Data Science

Saint Petersburg State University, Russia

Sep. 2018 – June 2022

Bachelor in Computer Science, Faculty of Math and Computer Science

GPA 4.91/5.0 (with Honors)

PROFESSIONAL EXPERIENCE

Oracle Labs, Research Intern

Aug. 2025 – Present

LLMs & AI Agents for Schema Mapping

Zurich, Switzerland

- Working on LLM- and agent-based approaches for automated schema mapping; responsible for configuring and evaluating schema enriching pipelines.

JetBrains, ML Engineer

Sep. 2022 – Sep. 2024

JetBrains AI Team

Paphos, Cyprus

- Contributed to Mellum LLM and code completion system as part of the Evaluation team; owned A/B testing metrics, reporting, and data processing.
- Developed a unified A/B testing pipeline across multiple languages and IDEs, enabling systematic comparison of completion models.
- Trained and deployed CatBoost models for IDE search ranking, improving key ranking metrics by 15%.
- Optimized metric/log collection algorithms for online experiments, reducing storage usage by 30%.
- Implemented features for documentation generation and intelligent refactoring suggestions in the JetBrains AI Assistant plugin.

Huawei Research, Junior ML Engineer

May 2022 – Aug. 2022

ML Research Team

Saint Petersburg, Russia

- Contributed to a question-answering tool for programming tasks using large language and code models.
- Designed a code-centric document retrieval system, experimenting with DPR, RAG, and Transformer-based encoders.
- Improved Python documentation search, doubling PR@1 over a strong ElasticSearch baseline.

JetBrains Research, Intern

Sep. 2020 – May 2022

Machine Learning for Software Engineering Lab

Saint Petersburg, Russia

- Developed code-comment inconsistency detection plugin, using LSTM-based models;
- Constructed a large open-source dataset (4000+ Java projects) and refined it with a custom crowdsourcing and active learning pipeline.

PROJECTS

Post-training for SwissAI (Apertus): SFT → DPO Reproduction

Feb. 2025 – Jun. 2025

Master Project Student, CLAIRE Lab (EPFL)

- Reproduced the Tülu3 post-training pipeline on Llama 3.1 8B and adapted it to SwissAI **Apertus** (8B/70B): Supervised Fine-Tuning (SFT) followed by Direct Preference Optimization (DPO).
- Extended evaluation infrastructure by forking AllenAI's `olmes` (github.com/swiss-ai/olmes); added vLLM inference with tensor parallelism, standard benchmarks (MMLU, GSM8K, HumanEval, IFEval, BBH, TruthfulQA, PopQA), and automated SLURM job generation.
- Implemented and productionized a TRL/DeepSpeed-based DPO trainer in the SwissAI codebase (github.com/swiss-ai/swiss-alignment); ran a β -sweep $\{0.01, 0.05, 0.1\}$ with full W&B logging and reproducibility scripts.

EDUCATION & VOLUNTEERING

Teaching Assistant | Neapolis University Pafos, Cyprus

Oct. 2023 – Sep. 2024

- Collaborated with Prof. Alexander Kulikov on Discrete Mathematics I: exercise sessions, grading, and problem preparation.

Instructor | Olympiad Mathematics Club for Schoolchildren, Paphos, Cyprus

Sep. 2022 – May 2023